

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Discrete Structure**

Code No: **CSC165** Semester: **2nd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. How can you use mathematical induction to prove statements? Use mathematical induction to show that the sum of first n positive integers is $n(n+1)/2$.
2. Explain linear homogeneous recurrence relation with constant coefficients. What is the solution of the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1$ and $a_1 = 6$?
3. What is shortest path problem? Use Dijkstra's shortest path algorithm to find the shortest path between the vertices a and z in the weighted graph given below.

Section B

Attempt any Eight questions[8×5=40]

4. Let us assume that R be a relation on the set of ordered pairs of positive integers such that $((a,b), (c,d)) \in R$ if and only if $ad=bc$. Is R an equivalence relation?
5. Define function. Let f_1 and f_2 be functions from R to R such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?
6. Explain fuzzy set with example. How do you find complement of a fuzzy set?
7. What is congruent modulo? Determine whether 37 is congruent to 3 modulo 7 and whether -29 is congruent to 5 modulo 17.
8. Define network flow with example. What are saturated edge, unsaturated edge and slack value?
9. Give an example of tautology and contradiction. Show that implication and contrapositive are equivalence.
10. What is direct proof? Give a direct proof that if m and n are both perfect squares, then mn is also a perfect square.
11. What is product rule? How many strings are there of four lowercase letters that have the letter x in them?
12. Explain matrix representation of relations with example.